

속도는벡터 · 소개영상 explainer animation — Veo 3.1 16 segment 정보 (v3)

26-1 인공지능종합설계 · 캡스톤 디자인 · 연세대학교 BDAI 연구실

작성 2026-05-27 14:46 KST · 5/28 12:00 LearnUs 마감 약 21 시간 남음

★ 본 영상 = ref 영상 ([submission/_drafts/archive/2026_05_26_postsubmit/video/무제.mp4](#) , 31.8초 무음 explainer)
base + Act 2·3·4 확장. 사용자 명시 5/27 14:00 KST — "공상학적 cinematic X · 실제 알고리즘 동작 explainer animation · 발표 슬라이드 첨부 영상 느낌 살리기 · 쿼리 추출 → 우리 연구까지 narrative".

★ 이전 cinematic 19-segment storyboard ([속도는벡터_소개영상_Veo3_1_cinematic_19segment_20260527_134000.md](#)) 폐기 — 본 문서가 새 정보.

0. 영상 사양

항목	값
총 길이	2:08 (128 초)
Segment	16 개 × 8 초 (Veo 3.1 segment)
해상도	1920 × 1080 (16:9 가로, 1080p)
프레임	30 fps (또는 24 fps cinematic)
오디오	Veo 3.1 native audio — 한국어 narration + 효과음 + clean ambient (cinematic 음악 X)
자막	영상 안 자막 없음 (Path A · 후처리 SRT)
비주얼 톤	흰 배경 + coral #D14F4F (부정확) ↔ cyan #3DAFC1 (정확) split · clean academic explainer · ref 영상 동일 톤
톤	clean academic · explainer animation · step-by-step algorithm reveal

1. 4 막 16 segment 구조

Act	시간	seg	hero	핵심 메시지
I 한 cell explainer (ref carry)	0:00-0:32	4 (1A·1B·1C·1D)	"7,599 행" → "7.24× 격차"	한 쿼리 → 한 추정 (33 vs 7,581) → 두 plan → 7,242 ms vs 1,000 ms
II 본 연구 설계	0:32-1:04	4 (2A·2B·2C·2D)	"한 곳만 바꾼다"	\$V-B\$ 베르누이 5 단계 → 단일 개입 (분포 인지) → 4 mode matched → 1,508 cell
III 결과	1:04-1:36	4 (3A·3B·3C·3D)	"89.1%" · "5.67×" · "94.9%"	결합 상위 + 음성 대조 (메커니즘 분리) + 엔진 plan 회복 + 비대칭 구조
IV 결론·팀	1:36-2:08	4 (4A·4B·4C·4D)	"negative + methodological"	기여 4 + honest limitation + 속도는벡터 + BDAI

2. Veo 3.1 prompt — 16 segment

각 prompt 는 Veo 3.1 (Ultra) Flow 에 직접 복붙. 영문 prompt + Korean narration.

Act I — 한 cell explainer (ref 영상 carry 재현)

Segment 1A · 0:00-0:08 — VECTOR RANGE QUERY 코드 + 의문

8-second clean academic explainer animation, 1920x1080, white background, minimal flat design.

VISUAL: A clean white background. In the center of the screen, a code block fades in with a monospace terminal-style font.

```
SELECT ps_partkey, ps_supplycost
FROM   partsupp_deep_10
WHERE  (a blinking cursor)
```

Below the code block, a centered question in Korean fades in: "이 쿼리는 몇 행을 반환할까?" with a small gray question mark icon.

CAMERA: Static framing, no camera movement. Subtle fade transitions for each element.

LIGHTING: Soft even white background. No dramatic lighting.

MOOD: Calm, academic, focused — a question is being posed.

KOREAN AUDIO NARRATION (calm academic male voice, slow pace): "벡터 술어를 포함한 분석 쿼리. PostgreSQL 플래너는 쿼리를 분석하고 최적의 실행 계획을 생성합니다."

SOUND EFFECTS: Soft typing sound as code appears. Subtle UI tick when the question appears.

MUSIC: Very subtle ambient pad in the background, low volume, no melody. No cinematic crescendos.

Segment 1B · 0:08-0:16 — GROUND TRUTH 7,599 행 reveal

8-second clean academic explainer animation, 1920x1080, white background, minimal flat design.

VISUAL: A clean white background. At the top, a small monospace text: "WHERE p_vec . . :q < 0.86". Below it, a large number "7,599" is revealed.

CAMERA: Static framing. Dots highlight in cyan sequentially with a wave effect. The "7,599" number reveals with a subtle animation.

LIGHTING: Soft even white background.

MOOD: Revealing, foundational — the ground truth.

KOREAN AUDIO NARRATION (calm academic male voice): "실제로 매칭되는 행 수는 칠천 오백 구십 구. 이 숫자를 정확히 집계합니다."

SOUND EFFECTS: Subtle dot tick sounds as dots highlight. A soft confirmation chime when 7,599 appears.

MUSIC: Very subtle ambient pad, low volume.

Segment 1C · 0:16-0:24 — PLANNER DEFAULT (33) vs ORACLE (7,581) split

8-second clean academic explainer animation, 1920x1080, white background, side-by-side comparison.

VISUAL: A clean white background split into two equal panels with a thin vertical divider. The header at the top reads "PLANNER DEFAULT (33) vs ORACLE (7,581) split".
Left panel (coral-red color scheme): A small red label "A · 부정확" at top, then "PLANNER DEFAULT" in gray monospace.
Right panel (cyan color scheme): A small cyan label "B · 정확" at top, then "ORACLE INJECTION" in gray monospace.

CAMERA: Static framing. Left and right panels reveal simultaneously with a horizontal sweep.

LIGHTING: Soft even white background.

MOOD: Contrast – two different views of the same query.

KOREAN AUDIO NARRATION (calm academic male voice): "플래너 기본 추정은 약 삼십 삼 행. 정답 카디널리티 주입은 칠천

SOUND EFFECTS: Two soft confirmation chimes – one for each panel.

MUSIC: Very subtle ambient pad, low volume.

Segment 1D · 0:24-0:32 — 실행 + 7,242 ms vs 1,000 ms · 7.24× 격차 hero

8-second clean academic explainer animation, 1920x1080, white background, side-by-side comparison + final number reveal.

VISUAL: A clean white background. First half (0–4s): Side-by-side execution panels. Left (coral): "× PGVECTOR 기본 추정은 약 삼십 삼 행. 정답 카디널리티 주입은 칠천 이백 사십". Right (cyan): "× ORACLE INJECTION 기본 추정은 약 삼십 삼 행. 정답 카디널리티 주입은 칠천 이백 사십".
Second half (4–8s): The two panels shrink to the top edge. In the center of the screen, a final hero reveal – mass difference.

CAMERA: Static framing. The two progress bars animate at different speeds. Numbers reveal with smooth scale-up.

LIGHTING: Soft even white background.

MOOD: Decisive – the cost of one mis-estimation.

KOREAN AUDIO NARRATION (calm academic male voice): "같은 쿼리, 같은 엔진, 같은 하드웨어에서 결과는 칠천 이백 사십

SOUND EFFECTS: Slow grinding sound for the red bar. Quick swoosh for the cyan bar. Final chime for "7.24×" reveal.

MUSIC: Very subtle ambient pad, slight emphasis at the final reveal.

Act II — 본 연구 설계

Segment 2A · 0:32-0:40 — "이 한 숫자를 어떻게 더 잘 짚을 것인가?"

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a large bold question in Korean: "이 한 숫자를 어떻게 더 잘 짚을

CAMERA: Static framing. Question fades in. Then the diagram slides in from below.

LIGHTING: Soft even white background.

MOOD: Pivoting from problem to method – the research begins.

KOREAN AUDIO NARRATION (calm academic male voice): "이 한 숫자를 어떻게 더 잘 짚을 것인가. 한 논문이 제시한 적응적

SOUND EFFECTS: Soft paper sound when the paper box appears. Subtle ticks as each stage of the diagram appears.

MUSIC: Very subtle ambient pad, low volume.

Segment 2B · 0:40-0:48 — 단일 개입 (베르누이 → 분포 인지 stratification) zoom in

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. The 5-stage horizontal flow diagram from previous segment is shown at the top, s

CAMERA: Static framing with a dramatic push-in on the first circle as it expands. Arrow animates from left to right

LIGHTING: Soft even white background.

MOOD: Pivotal – the single intervention.

KOREAN AUDIO NARRATION (calm academic male voice): "본 연구는 표본 선택 단계 한 곳만 분포 인지 층화 표본 추출로 바

SOUND EFFECTS: A whoosh sound as the camera zooms in. A confirmation chime when the cyan arrow lands.

MUSIC: Very subtle ambient pad, slight emphasis at "한 곳만".

Segment 2C · 0:48-0:56 — 4 mode matched 측정 (베이스라인·단독 대체·결합·정답)

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a small black circle labeled "한 측정 셀" appears. From this center, four arrows point outwards to four colored boxes.

Top-left box (gray): "기본 엔진 · 고정 비율 추정"

Top-right box (red): "베이스라인 · 무작위 베르누이 (논문 그대로)"

Bottom-left box (yellow-orange): "단독 대체 · method 단독 (음성 대조군)"

Bottom-right box (cyan): "결합 · 베이스라인 + 분포 인지 평균"

Each arrow appears one by one in sequence. Below the diagram: "한 측정 셀에서 네 방식이 동시에 산출 · matched."

CAMERA: Static framing. The 4 arrows and boxes appear sequentially with subtle fade animations.

LIGHTING: Soft even white background.

MOOD: Structural reveal – the controlled experimental design.

KOREAN AUDIO NARRATION (calm academic male voice): "한 측정 셀에서 네 방식을 동시에 산출합니다. 기본 엔진, 베이스라인, 단독 대체, 그리고 결합 방식이 모두 산출됩니다."

SOUND EFFECTS: 4 soft "appear" chimes – one per box.

MUSIC: Very subtle ambient pad.

Segment 2D · 0:56-1:04 — 1,508 cell 측정 평면

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a 3D isometric grid materializes – a transparent rectangular box filled with small cubes.

CAMERA: Slow 360-degree rotation around the 3D grid for cinematic effect.

LIGHTING: Soft even white background. Subtle cyan glow from the cubes.

MOOD: Scale and rigor – the measurement plane.

KOREAN AUDIO NARRATION (calm academic male voice): "5 데이터셋, 5 조작변인, 13 method 의 천 오백 팔 측정 셀 전수에 걸쳐 측정 평면이 생성됩니다."

SOUND EFFECTS: A rapid sequence of soft ticks as the cubes fill.

MUSIC: Very subtle ambient pad.

Act III — 결과

Segment 3A · 1:04-1:12 — hero "89.1%" + paired $\Delta\%$ -4.38%

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a massive bold sans-serif number reveals with a smooth scale-up:

CAMERA: Static framing. Number reveals with smooth scale-up. Histogram bars rise in sequence from left to right.

LIGHTING: Soft even white background.

MOOD: Dramatic result reveal – the headline finding.

KOREAN AUDIO NARRATION (calm academic male voice): "1,508 측정 모두에서 짝지어 비교했습니다. 결합 방식이 89.1 퍼센

SOUND EFFECTS: A soft confirmation chime when "89.1%" appears. Subtle ticks for histogram bars.

MUSIC: Very subtle ambient pad, slight emphasis at the number reveal.

Segment 3B · 1:12-1:20 — hero "35.2%" 음성 대조군 + 메커니즘 = 앙상블 효과

8-second clean academic explainer animation, 1920x1080, white background, comparison reveal.

VISUAL: A clean white background. Left side: A small label "단독 대체" in muted yellow-orange tone, followed by a

CAMERA: Static framing. Left side appears first, then " $\neq$ ", then right side. The insight at the bottom fades in last.

LIGHTING: Soft even white background.

MOOD: Critical contrast – the mechanism revealed.

KOREAN AUDIO NARRATION (calm academic male voice): "그런데 단독 대체 방식은 35.2 퍼센트에서만 이깁니다. 동전 던지

SOUND EFFECTS: Soft chime for "35.2%", a low click for " $\neq$ ", a clear confirmation chime for the insight reveal.

MUSIC: Very subtle ambient pad.

Segment 3C · 1:20-1:28 — hero "5.67x" 엔진 가속 + 12 cell heatmap

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top half: A horizontal flow diagram showing "결합 추정값" arrow pointing to "pgv

CAMERA: Static framing. Number reveal first, then heatmap cells fill in sequence.

LIGHTING: Soft even white background.

MOOD: Engineering result – speedup is real.

KOREAN AUDIO NARRATION (calm academic male voice): "결합 추정값을 pgvector 엔진에 직접 주입했습니다. 12 셀 평균 5.

SOUND EFFECTS: Soft chime for "5.67x". Sequential ticks as heatmap cells fill.

MUSIC: Very subtle ambient pad.

Segment 3D · 1:28-1:36 — hero "94.9%" plan 회복 + 4 case 매트릭스 + 비대칭

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Center top: A massive bold number "94.9%" in cyan, with a smaller subscript "결합

Top-left (cyan, TP): "TP · 90" with caption "둘 다 정답"

Top-right (light red, FN): "FN · 1" with caption "결합만 망친 (Q3 sparse_rp 1건)"

Bottom-left (cyan-highlighted, FP): "FP · 58 ★" with caption "결합이 회복한 cell"

Bottom-right (gray, TN): "TN · 7" with caption "둘 다 오답"

The FP cell glows brightest, highlighted with a thin cyan border. Below the matrix: a clear short insight: "결합 회

CAMERA: Static framing. The 4 matrix cells reveal in sequence – TP first, then FN, then the highlighted FP, then T

LIGHTING: Soft even white background.

MOOD: Structural advantage – the asymmetric win.

KOREAN AUDIO NARRATION (calm academic male voice): "결합 13종이 정답 plan 을 94.9 퍼센트 회복합니다. 베이스라인이

SOUND EFFECTS: A confirmation chime for "94.9%", 4 distinct chimes for each matrix cell, with the brightest chime

MUSIC: Very subtle ambient pad.

Segment 4A · 1:36-1:44 — 본 연구 기여 4

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. In the center, a horizontal arrangement of 4 small academic-style cards, each with an icon and a label.

Card 1: A small "fork-arrow" icon → label "메커니즘 분리"

Card 2: A small "balance-scale" icon → label "음성 대조군"

Card 3: A small "stop-sign" icon → label "구조적 한계 규명"

Card 4: A small "gear-fast" icon → label "엔지니어링 5.67x 가속"

Each card reveals in sequence from left to right. Below the cards, a centered emphasized text: "negative + methodological limitations"

CAMERA: Static framing. Cards reveal one by one.

LIGHTING: Soft even white background.

MOOD: Closing summary – what we contributed.

KOREAN AUDIO NARRATION (calm academic male voice): "본 연구의 기여 네 가지. 메커니즘 분리, 음성 대조군, 구조적 한계 규명, 엔지니어링 가속."

SOUND EFFECTS: 4 soft "appear" chimes – one per card.

MUSIC: Very subtle ambient pad.

Segment 4B · 1:44-1:52 — honest limitation 정직 보고

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top center: A small header "honest limitation · 정직하게 남기는 한계". Below, a list of 4 bullet points.

"• 다중 벡터 864 차원 군집화 파탄 – 2 건 이상치 제외"

"• method 명칭 정직성 – 8 method PCA alias 정정"

"• 통계 검정 floor $p = 1/1024$ – 효과크기로 판단"

"• 엔진 latency 동등 86.9% small effect – 구조적 한계"

Each bullet point fades in one by one. Below the list: "학술적 정직성 자체가 본 연구의 contribution 가운데 하나."

CAMERA: Static framing. Bullets reveal sequentially.

LIGHTING: Soft even white background.

MOOD: Honest, transparent – no over-claiming.

KOREAN AUDIO NARRATION (calm academic male voice): "본 연구의 한계도 정직하게 남깁니다. 다중 벡터 군집화 파탄, method 명칭 정직성, 통계 검정 floor, 엔진 latency 동등."

SOUND EFFECTS: 4 small ticks as bullets appear.

MUSIC: Very subtle ambient pad.

Segment 4C · 1:52-2:00 — 속도는벡터 팀 + 지도진

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Top center: A large bold team name "속도는벡터" in navy color. Below the team name

CAMERA: Static framing. Team name reveals first, then team avatars, then mentors.

LIGHTING: Soft even white background.

MOOD: Warm team reveal.

KOREAN AUDIO NARRATION (calm academic male voice): "속도는벡터 팀. 박세은, 강재현, 조현빈, 이동욱. 지도해 주신 박

SOUND EFFECTS: A warm confirmation chime when the team name appears.

MUSIC: Very subtle ambient pad with a slight warm resolution.

Segment 4D · 2:00-2:08 — 감사합니다 + 마무리

8-second clean academic explainer animation, 1920x1080, white background.

VISUAL: A clean white background. Center: A large bold text in Korean: "감사합니다." Below it, a smaller text: "속

CAMERA: Static framing. Smooth fade in for the closing text, slow fade out at the very end.

LIGHTING: Soft even white background.

MOOD: Warm, calm closing.

KOREAN AUDIO NARRATION (calm academic male voice, warm tone): "감사합니다."

SOUND EFFECTS: A warm closing chime, then ambient silence.

MUSIC: Very subtle ambient pad resolves to silence at the end.

3. 합성 workflow

3.1 Flow 진행 (Vevo 3.1 Quality)

1. <https://labs.google/fx/ko/tools/flow> → 기존 프로젝트 "속도는벡터 캡스톤 소개영상" 진입
2. 우측 chat 입력창에 위 §2 의 Segment 1A → 4D prompt 16 개 순차 복붙
3. 각 prompt 당 약 60-120s generation (Vevo 3.1 Quality)
4. 각 segment 출력 mp4 확인 + 다운로드
5. 기존 cinematic 1A·1B·1C 영상은 무시 (Flow 의 동영상 라이브러리에서 자동 삭제 또는 무시)

3.2 합성

옵션 A: Flow 자체 Storyboard 모드 (추천) - 좌측 "장면" tab → 새 storyboard → Act 1A-4D 순서대로 16 segment 배치 - segment 간 transition = **fade through white 0.2s** (clean academic 톤) - Export → 1080p mp4 (128초)

옵션 B: DaVinci Resolve - 16 mp4 import → timeline 순서 배치 - transition Cross Dissolve 0.2s - (선택) 자막 burn-in SRT 추가 - Apple SD Gothic Neo 한국어 자막 - Deliver → YouTube 1080p preset

3.3 자막 burn-in 후처리 (Path A carry)

영상 안 한국어 자막 없음 (Veo 한글 깨짐 회피). 합성 단계에서 SRT 별도 추가: - 각 segment 의 KOREAN AUDIO NARRATION 텍스트 = 자막 텍스트 - timing = segment 시작-종료 (8초 단위) - 폰트 Apple SD Gothic Neo, 흰 텍스트 + navy **#1E3A5F** 그림자, 화면 하단 중앙

3.4 YouTube + QR + 포스터

1. YouTube Unlisted 업로드 (제목·설명 carry 메모)
2. URL → qr-code-generator.com PNG (500×500)
3. 포스터 우측 footer QR placeholder 갱신 → PDF re-export → LearnUs 5/28 12:00 마감 제출

4. ref 영상 frame 분석 carry (시각 디테일 참고)

Frame	시간	본 영상 매핑	핵심 시각
1	0-4s	Segment 1A	VECTOR RANGE QUERY code + "이 쿼리는 몇 행을 반환할까?"
2	4-8s	Segment 1B	도트 매트릭스 + "7,599 행" GROUND TRUTH
3	8-12s	Segment 1C 좌·우 split	"≈ 33 행" coral vs "= 7,581 행" cyan
4	12-16s	Segment 1C plan tree	plan tree 비교 (Hash Join vs Index+NL)
5	16-20s	Segment 1D 실행 시작	CPU·MEM·ELAPSED 0 ms
6	20-24s	Segment 1D 진행 중	좌 3,828 ms 진행 / 우 1,000 ms 완료
7	24-28s	Segment 1D 완료	좌 7,242 ms 완료 / 우 1,000 ms 완료
8	28-32s	Segment 1D hero	"7.24x 격차" + "추정 1건 → plan 1개 → 자원 7배"

5. 핵심 정량 수치 (Veo prompt 안 직접 사용)

위치	수치	의미
Segment 1B	7,599 행	한 cell ground truth
Segment 1C	33 vs 7,581 행	planner default vs oracle
Segment 1D	7,242 ms vs 1,000 ms · 7.24x	한 cell latency 격차 (ref 영상 동일)
Segment 2D	1,508 cell	5 데이터셋 × 5 조작변인 × 13 method
Segment 3A	89.1%	1,344/1,508 cell better (paired Δ% 중앙값 -4.38%)
Segment 3B	35.2%	단독 대체 better (음성 대조군)
Segment 3C	5.67x	12 cell 평균 엔진 가속
Segment 3D	94.9% · 89.2% · 1.1%	plan 회복 (148/156) · 회복률 · 망친 비율

6. 환각 회피 룰 (carry · 변경 X)

- 코드명 노출 금지 (B1·CaseA·CaseB·oracle·baseline)
- 한국어 라벨 통일 (기본 엔진·베이스라인·단독 대체·결합·정답)
- 영상 안 자막 X (Veo 한글 깨짐 회피 · Path A 후처리 SRT)
- ref 영상 톤 carry — 흰 배경 + coral/cyan split + 도트 매트릭스 + plan tree + CPU/MEM meter

작성 2026-05-27 14:46 KST · 5/28 12:00 LearnUs 영상·포스터 마감 약 21 시간 남음 · 16 segment explainer animation 2:08 · ref 영상 (**무제.mp4** 31.8s) base + Act 2·3·4 확장 · 사용자 명시 (공상학적 X · 실제 알고리즘 동작 explainer animation · 발표 슬라이드 첨부 영상 느낌).